

Surname

Name

University ID Number

Signature

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- Write the solutions (including procedures and intermediate steps) in the blank areas (use the back of the page, if needed)
 - The number of pages is 8. Additional papers will not be considered.
 - Use of books, lecture notes or any other material is forbidden. Electronics (smartphones, calculators, etc.) are forbidden.
 - Clarity, order and precision are strongly considered for the final evaluation.
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1. [Linear classification]

Consider a perceptron with weight vector $(w_0, w_1, w_2) = (2, 1, 1)$.

- Plot the partition of $\mathbb{R}^2$ that is realized by this perceptron in a diagram and mark the area where the perceptron outputs 1.
- Which of the perceptrons with the following weight vectors have the same hyperplane and which ones represent exactly the same classification (i.e., target prediction) as the perceptron given above?

$$P1: (w_0, w_1, w_2) = (1, 0.5, 0.5)$$

$$P2: (w_0, w_1, w_2) = (200, 100, 100)$$

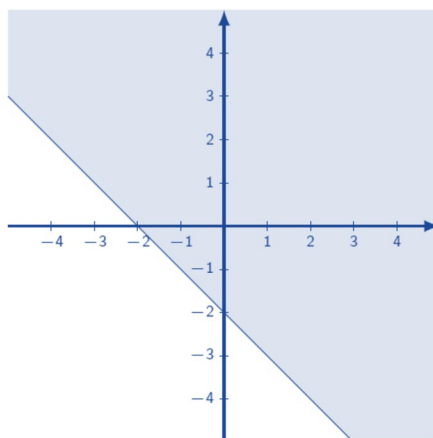
$$P3: (w_0, w_1, w_2) = (\sqrt{2}, 1, 1)$$

$$P4: (w_0, w_1, w_2) = (-2, -1, -1)$$

- Describe one way to train a perceptron using a set of supervised data.

SOLUTIONS:

a.



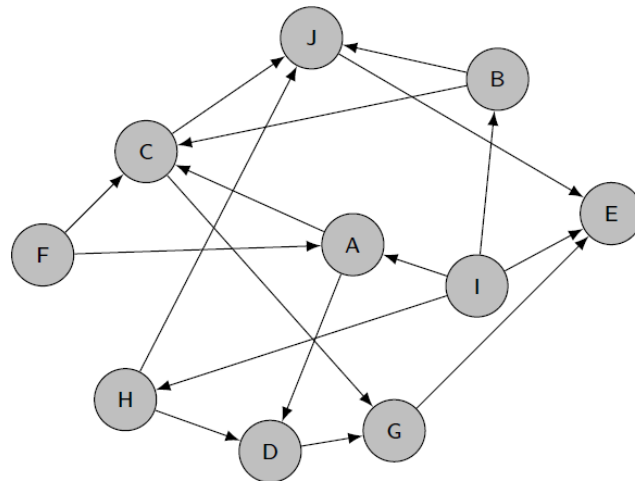
b.

(w_0, w_1, w_2)	Same hyperplane	Same classification
$(1, 0.5, 0.5)$	X	X
$(200, 100, 100)$	X	X
$(\sqrt{2}, 1, 1)$		
$(-2, -1, -1)$	X	

- See the lecture notes (chapter 02).

2. [Neural networks]

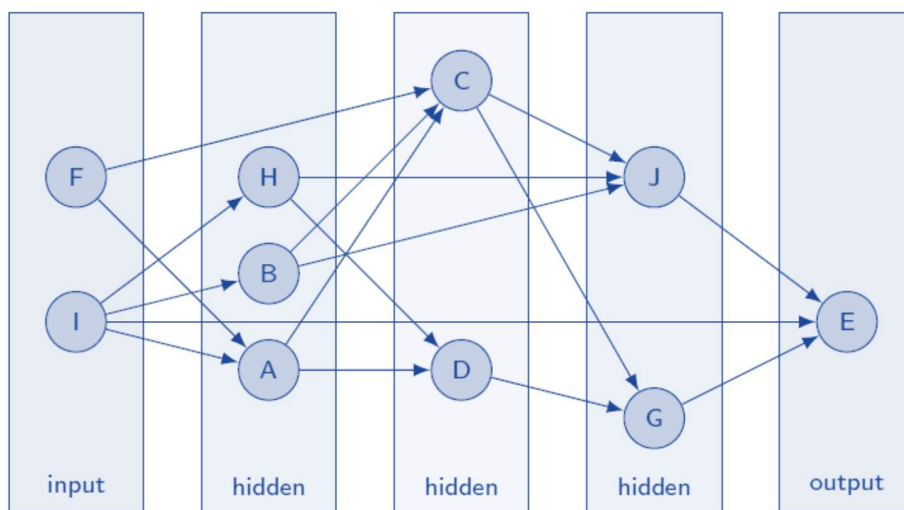
Consider the network structure of a multi-layer perceptron with 10 neurons given in the figure below. Each circle denotes a neuron, the arrows denote connections between neurons.



- Analyze the structure of the network: which of the neurons are input neurons, which ones are output neurons? How many layers does this network have? Which neurons belong to which layer?
- Describe the backpropagation algorithm for training feedforward neural networks in case of logistic activation functions.

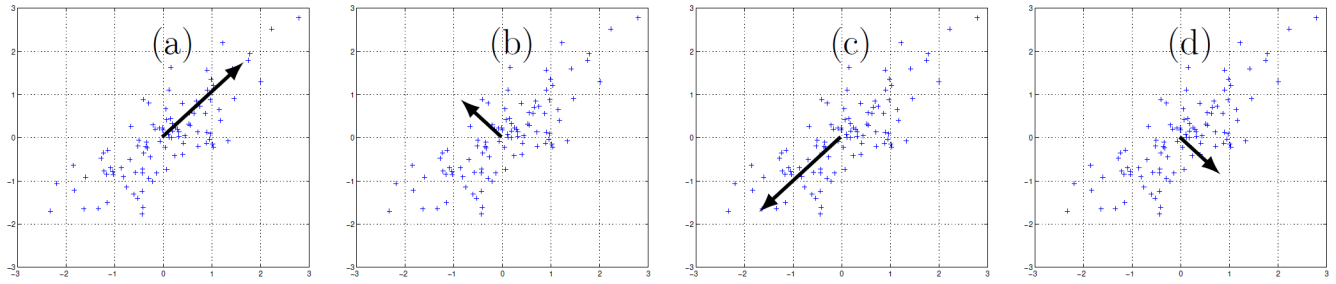
SOLUTIONS:

- After re-ordering, the following scheme (answering to all questions) arises:



- See the lecture notes (chapter 09).

3. [PCA]



- Which of the above figures correspond to possible values that PCA may return for the first eigenvector/principal component? Why?
- Explain why SVD can be seen as an effective tool for dimensionality reduction even in case of noisy data.

SOLUTIONS:

- (a) The maximal variance is along the $y = x$ line, so this option is correct.

(b) The first principal component is aligned with the direction of maximal variance, whereas this is aligned with the direction of minimal variance.

(c) The maximal variance is along the $y = x$ line, so the negative vector along that line is correct for the first principal component.

(d) See again (b).
- See the lecture notes (chapter 11).

4. [Mathematical foundations of learning]

Consider the linear regression problem.

- a. Explain why optimizing the fitting error does not necessarily minimize the generalization error.
- b. Show that the impact of the measurement noise on the out-of-sample error does not depend on choice of the hypothesis set, if the noise and the input are uncorrelated.
- c. In order to estimate the out-of-sample prediction performance, the dataset available for learning is usually divided in three subsets: training, validation and testing. Why? Which is the role of each set?

SOLUTIONS:

- a. See the lecture notes (chapter 1).
- b. See the lecture notes (chapter 7).
- c. See the lecture notes (chapter 8).

5. [Analysis and prediction of ARMA/ARMAX processes]

Consider the process $y(t)$, defined as

$$y(t) + \frac{2}{3}y(t-1) = x(t)$$

$$x(t) = w(t) + v(t)$$

$$w(t) = e(t) + \frac{1}{3}e(t-1)$$

with $e \sim wn(0,9)$ and $v \sim wn(0,1)$ being uncorrelated noises.

- Is $y(t)$ stationary?
- Compute the mean m_y and the covariance $\gamma_y(\tau)$ of the process.
- Find the spectrum $\Gamma_y(\omega)$ of $y(t)$.
- Compute the optimal 1-step ahead predictor $\hat{y}(t|t-1)$.

SOLUTIONS:

- As sum of stationary processes, the process is stationary.
- The mean $m_y = 0$ since the process corresponds to the sum of zero-mean stationary processes. The covariance can be found as:

$$\gamma_y(\tau) = -\frac{2}{3}\gamma_y(\tau-1) + E[e(t)y(t-\tau)] + \frac{1}{3}E[e(t-1)y(t-\tau)] + E[v(t)y(t-\tau)]$$

Due to the features of the process it holds that

- $E[e(t)y(t)] = 9$, while $E[e(t)y(t-\tau)] = 0$ for all τ such that $\tau \neq 0$
- $E[e(t-1)y(t-\tau)] = -3$ when $\tau = 0$, $E[e(t-1)y(t-\tau)] = 9$ if $|\tau| = 1$, and it is zero otherwise
- $E[v(t)y(t-\tau)] = 1$ if $\tau = 0$, and zero otherwise.

Through these conditions the covariance satisfies:

- $\gamma_y(0) = \frac{63}{5}$
- $\gamma_y(1) = -\frac{27}{5}$
- $\gamma_y(\tau) = -\frac{2}{3}\gamma_y(\tau-1)$, for all τ such that $|\tau| > 1$.

- The spectrum can be computed as the sum of the spectra of the two stationary process constituting $y(t)$. In particular:

$$\Gamma_y(\omega) = \frac{99 + 54 \cos(\omega)}{13 + 12 \cos(\omega)}$$

- Let $y_1(t)$ and $y_2(t)$ be the two stationary processes such that $y(t) = y_1(t) + y_2(t)$, then the optimal 1-step predictor is defined as

$$\hat{y}(t|t-1) = -\frac{1}{3} \frac{1}{1 + \frac{1}{3}z^{-1}} \hat{y}_1(t|t-1) - \frac{2}{3} \hat{y}_2(t|t-1)$$

6. [Kalman filter]

Consider the system

$$x_1(t+1) = -\frac{1}{3}x_1(t) + v(t)$$

$$x_2(t+1) = \frac{1}{2}x_2(t) + 2w(t)$$

$$y(t) = x_2(t) + w(t)$$

with $w \sim wn(0,1)$ and $v \sim wn(0,1)$ being uncorrelated noises.

- Is the variance of the steady state prediction error $x(t) - \hat{x}(t|t-1)$ bounded?
- Draw the block scheme comprising the system and the asymptotic Kalman filter.
- Compute the variance of the innovation $e(t) = y(t) - \hat{y}(t|t-1)$.

SOLUTIONS:

- The system can be split into two subsystem, respectively given by:

- Subsystem A: $x_1(t+1) = -\frac{1}{3}x_1(t) + v(t); y_A(t) = 0$
- Subsystem B: $x_2(t+1) = \frac{1}{2}x_2(t) + 2w(t); y_B(t) = x_2(t) + w(t)$

Subsystem A is stable and $V_{1,2} = 0$. Therefore, the first asymptotic theorem can be applied and, thus, the prediction error $x_1(t) - \hat{x}_1(t|t-1)$ is bounded. In particular, $K_1(t) = 0$, for all t .

Subsystem B is still stable, but $V_{1,2} = E[2w(t)w(t)] = 2 \neq 0$. Therefore, none of the asymptotic theorems can be applied.

For this subsystem we study the DRE: $P_2(t+1) = \frac{\frac{9}{4}P_2(t)}{P_2(t)+1}$ (Vertical asymptote: $P_2(t) = -1$; Horizontal asymptote: $P_2(t+1) = \frac{9}{4}$)

We then check the solutions of the ARE: $\bar{P}_2^1 = 0$ and $\bar{P}_2^2 = \frac{5}{4}$

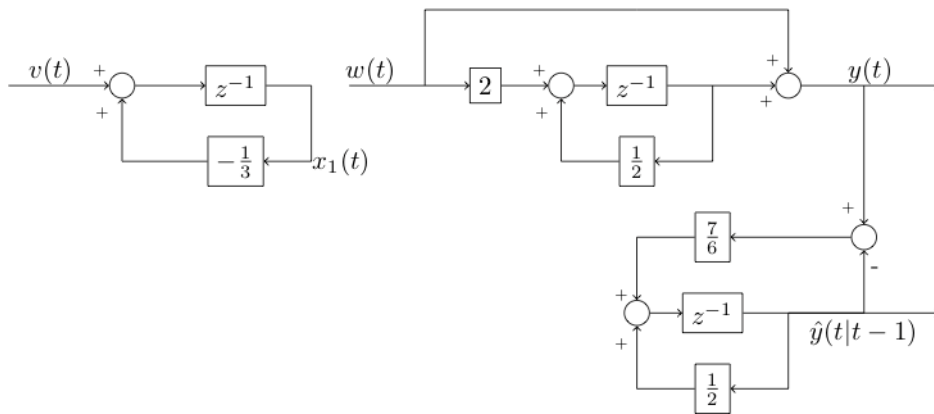
It can be proven that the DRE:

- Stays in 0 if $P_2(0) = 0$. This corresponds to a Kalman gain equal to zero
- Converges to $\bar{P}_2^2$ if $P_2(0) \neq 0$. This corresponds to $K_2 = \frac{7}{6}$

When $P_2(t)$ stays in zero, then the Kalman gain is a zero vector and, since the system is stable the prediction error is bounded. When $P_2(t)$ converges to $\bar{P}_2^2$ then $F_B - K_2H_B = \frac{1}{2} - \frac{7}{6} = -\frac{2}{3} \in (-1,1)$. Therefore, the prediction error is still bounded.

- See below.

- $var(e(t)) = var(x_2(t) - \hat{x}_2(t|t-1) + w(t)) = P_2(t) + 1$



7. [State-Space identification]

Consider the state-space (SS) representation of a discrete-time Linear Time Invariant system. Starting from this representation:

- a) Find the formula to obtain the Transfer-Function representation
- b) Find the formula to obtain the Impulse-Response representation

[See lecture notes \(theory\)](#)

8. [Frequency-domain identification]

Describe in detail the “step 1” (experiment design and pre-processing) of a frequency-domain system identification method.

[See lecture notes \(theory\)](#)